Category
Interest Rate Risk and Return

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SOURCES OF RETURN FROM INVESTING IN A FIXED-RATE BOND

Learning Outcome

- calculate and interpret the sources of return from investing in a fixed-rate bond;

Fixed-rate bond investors have three sources of return: (1) receipt of promised coupon and principal payments on the scheduled dates, (2) reinvestment of coupon payments, and (3) potential capital gains or losses on the sale of the bond prior to maturity. We now focus on how interest rate changes affect the reinvestment of coupon payments and a bond's market price if sold *prior to* maturity.

We start with two investors buying BRWA's new 10-year, 6.2% (annual) coupon eurobond, but each investor has a different time horizon for holding it. Initially, interest rates are unchanged. Then, we demonstrate the impact of higher interest rates on investors' total return. Finally, we show the effect of lower interest rates.

EXAMPLE 1

The first investor, Viswan Family Office (VFO), is a "buy-and-hold" investor, purchasing BRWA's new 10-year, 6.2% (annual) coupon eurobond priced at par (settlement: 15 October 2025; maturity: 15 October 2035) and holds it to maturity. If the coupon payments are reinvested at 6.2%, the yield-to-maturity, the future value of the coupons on the bond's maturity date, is found using Microsoft Excel's FV function:

= -*FV*(*Rate*, *Nper*, *Pmt*, *Pv*, *Type*),

where *Type* indicates payment at the end (0) or beginning (1) of the period. For this bond,

= -*FV*(0.062, 10, 6.2, 0, 0) = 82.493 per 100 of par value.

While 62 (= 6.2 × 10) in total coupons is received, the excess amount, 20.493 (= 82.493 – 62), is the "interest-on-interest" gain from reinvesting and compounding. The investor's rate of return, expressed as a compound annual growth rate, is 6.20%.

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{82.493 + 100}{100}\right)^{\frac{1}{10}} - 1.$$

r = 0.0620 = 6.20%.

Example 1 demonstrates that yield-to-maturity at the time of purchase equals the investor's rate of return under three assumptions: (1) the investor holds the bond to maturity, (2) there is no default by the issuer, and (3) the coupon interest payments are reinvested at that same rate of interest.

Example 2 considers the second investor, Baywhite Financial (Baywhite), which also buys the 10-year, 6.2% annual coupon payment bond and pays the same price. However, Baywhite has a four-year investment horizon, so coupon interest is reinvested for four years, and the bond is sold immediately after receiving the fourth coupon payment.

coupon payments are reinvested at 6.2% for four years, the future value of the reinvested coupons is 27.20321 per 100 of par value.

= −*FV*(0.062, 4, 6.2, 0, 0) = 27.203.

Interest-on-interest gain from reinvesting and compounding is 2.403 (= 27.203 – 24.800). When the bond is sold, it has six years remaining to maturity. Assuming yield-to-maturity remains 6.2%, the sale price is still 100% of par.

Baywhite's annualized rate of return is 6.20%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{27.203 + 100}{100}\right)^{\frac{1}{4}} - 1.$$

r = 0.0620 = 6.20%.

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Horizon yield is an investor's IRR based on the total return (reinvested coupon payments plus sale price or redemption amount) and the purchase price of the bond. The horizon yield on a bond investment is the annualized holding-period rate of return.

In Example 2, Baywhite's horizon yield is 6.2%, which matches the original yield-to-maturity even though Baywhite did not hold the bond until maturity. This example shows that the realized horizon yield matches the original yield-to-maturity if (1) coupon payments are reinvested at the same interest rate as the original yield-to-maturity and (2) the bond is sold at a price on the constant-yield price trajectory, implying there is no capital gain or loss when the bond is sold. Importantly, capital gains (losses) arise if a bond is sold at a price above (below) its constant-yield price trajectory, as shown in a prior lesson.

A point on the trajectory represents the **carrying value** of the bond at that time. The carrying value is the purchase price plus (minus) the amortized amount of the discount (premium) if the bond is purchased at a price below (above) par value. The constant-yield price trajectory for the investors in Example 1 and Example 2 is a flat curve at a price of 100 as the bonds were purchased and sold at par.

We now demonstrate the impact of *higher* interest rates on VFO and Baywhite by increasing interest rates by 100 basis points (bps) immediately after investment. The yield on the bond increases from 6.20% to 7.20%, and coupon reinvestment rates go up by 100 bps as well.

EXAMPLE 3

VFO, the buy-and-hold investor, purchases BRWA's new 10-year, 6.2% (annual) coupon eurobond at par, and then interest rates immediately increase to 7.20%. The future value of the reinvested coupons at 7.20% for 10 years is

= −*FV*(0.072, 10, 6.2, 0, 0) = 86.475.

And VFO's realized rate of return (horizon yield) is 6.43%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{86.475 + 100}{100}\right)^{\frac{1}{10}} - 1.$$

r = 0.0643 = 6.43%.

maturity.

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EXAMPLE 4

Baywhite buys the same BRWA eurobond at par, interest rates immediately rise to 7.2%, and it sells the bonds four years later. The future value of the reinvested coupons at 7.2% is

= -*FV*(0.072, 4, 6.2, 0, 0) = 27.609.

We must also calculate the sale price of the bond because the change in interest rates affects the bond's price and, unlike VFO, Baywhite is selling it before maturity. We calculate the sale price after four years as

= –*PV*(0.072, 6, 6.2, 100, 0) = 95.263,

resulting in a realized four-year horizon yield of 5.28%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{27.609 + 95.263}{100}\right)^{\frac{1}{4}} - 1.$$

r = 0.0528 = 5.28%.

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While Baywhite also benefits from the higher coupon reinvestment rate, it is more than offset by the capital loss from the sale of the bonds at the higher yield (lower price). Thus, Baywhite's rate of return is lower than in Example 2, where the interest rate remained at 6.2%.

We now examine the rates of return for these two investors if interest rates *decline* by 100 bps; the yield on the BRWA bonds fall from 6.20% to 5.20% immediately after they are purchased.

Knowledge Check

Immediately after VFO, the buy-and-hold investor, purchases BRWA's 10-year eurobonds at par, interest rates fall to 5.20%.

1. Calculate the future value of reinvested coupons.

Hide Solution

Solution:

The future value of reinvesting the coupon payments at 5.20% for 10 years is

= −*FV*(0.052, 10, 6.2, 0, 0) = 78.715.

2. Calculate VFO's horizon yield (realized rate of return).

Hide Solution

Solution:

VFO's horizon yield is 5.98%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

3. Contrast VFO's investment outcome in this scenario to Example 1 and Example 3.

Hide Solution

Solution:

VFO suffers from the lower coupon reinvestment rate. The realized horizon yield of 5.98% is 22 bps lower than the result in Example 1, when interest rates were unchanged, and 45 bps lower than the result in Example 3. There is no capital gain or loss, because the bonds are held until maturity. VFO's experience indicates that interest rate risk for a buy-and-hold investor arises entirely from reinvestment risk.

Baywhite buys BRWA's 10-year Eurobonds at par, and interest rates fall immediately to 5.20%. Baywhite sells the bonds after four years.

4. Calculate the future value of the reinvested coupons.

Hide Solution

Solution:

The future of the reinvested coupons is

$= -FV(0.052, 4, 6.2, 0, 0) = 26.802.$

5. Calculate Baywhite's horizon yield (realized rate of return).

Hide Solution

Solution:

The sale price of the bond after four years is

$= -PV(0.052, 6, 6.2, 100, 0) = 105.043,$

resulting in a realized four-year horizon yield of 7.16%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{26.802 + 105.043}{100}\right)^{\frac{1}{4}} - 1.$$

$r = 0.0716 = 7.16\%$

6. Contrast Baywhite's investment outcome in this scenario to Example 2 and Example 4.

Hide Solution

Solution:

The reduction in the future value of the coupons is more than offset by Baywhite's capital gain from the sale. Consequently, Baywhite's horizon yield is higher than in both Example 2 and Example 4.

As just demonstrated, interest income is the return associated with the *passage of time*, so it includes the receipt of coupon interest, the reinvestment of those coupons, and the amortization of the discount (or premium) from purchase at a price below (or above) par value to realign the return with the market discount rate. A capital gain or loss is the investor's return associated with the *change in the value* of the security, which for a fixed-

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investment horizon is at the heart of understanding interest rate risk and return. There are two offsetting types of interest rate risk that affect the bond investor: reinvestment risk and price risk. The future value of reinvested coupon payments *increases* when interest rates rise and *decreases* when rates fall. The sale price on a bond that matures after the horizon date *decreases* when interest rates rise and *increases* when rates fall.

Reinvestment risk matters more when the investor has a long-term investment horizon. A buy-and-hold investor only has reinvestment risk. Price risk matters more when the investor has a short-term investment horizon. An investor who sells before the first coupon is received has only price risk. Thus, two investors holding the same bond can have different exposures to interest rate risk if they have different investment horizons.

Question Set

1. *Fill in the blanks using the two words below:*

Reinvestment risk

Price risk

An investor who has a long-term investment horizon is more concerned about _____, while the investor with a short-term investment horizon is more concerned about _____.

Hide Solution

Solution:

An investor who has a long-term investment horizon is more concerned about reinvestment risk, while the investor with a short-term investment horizon is more concerned about price risk.

2. What is the carrying value of a bond?

Hide Solution

Solution:

The carrying value of a bond is the total value of the price at which the bond was purchased plus (minus) the adjustment for amortization of the discount (premium) if the bond is purchased at a price below (above) par value.

3. *Match the following*

Future value of coupons	Interest rates
A. Increase	Decrease
B. Decrease	Increase

Hide Solution

Solution:

The answer to A is increase. The future value of coupons has a direct relationship with interest rates. Rising interest rates mean coupons are reinvested at a higher rate, increasing their future value.

The answer to B is decrease. The future value of coupons has a direct relationship with interest rates. Falling interest rates mean coupons are reinvested at a lower rate, decreasing their future value.

investor's total return at the end of two years is *closest* to:

A. 5.23%.
B. 5.60%.
C. 5.75%.

Hide Solution

Solution:

A is correct. The investor's total return is composed of the (1) coupons, (2) reinvestment of coupons, and (3) gain/loss on sale prior to maturity.

The future value of the coupons assuming reinvestment at 5.90% is

= –*FV*(0.059, 2, 5.6, 0, 0) = 11.5304.

The sale price of the bond after two years, when three years remain to maturity, is

= –*PV*(0.059, 3, 5.6, 100, 0) = 99.1966.

The realized rate of return is 5.23%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{11.530 + 99.197}{100}\right)^{\frac{1}{2}} - 1.$$

r = 0.0523 = 5.23%.

B is incorrect because it is the rate of return if interest rates remain at 5.60%.

C is incorrect because it is the average of the two rates (i.e., the coupon rate, 5.6%, and the new, higher interest rate, 5.9%).

5. An investor purchases a seven-year, 6.7% (annual) coupon eurobond priced at par (settlement: 15 December 2031; maturity: 15 December 2038) and sells it after four years. Immediately after the purchase of the bond, interest rates fall to 6.3%. The investor's annualized rate of return at the end of four years is *closest* to:

A. 6.50%.
B. 6.66%.
C. 6.89%.

Hide Solution

Solution:

C is correct. The future value of the coupon and its reinvestment at 6.30% is

= –*FV*(0.063, 4, 6.7, 0, 0) = 29.441.

The sale price of the bond after four years is

= –*PV*(0.063, 3, 6.7, 100, 0) = 101.063.

The investor's rate of return is 6.89%:

$$r = \left(\frac{FV}{PV}\right)^{\frac{1}{T}} - 1.$$

$$r = \left(\frac{29.441 + 101.063}{100}\right)^{\frac{1}{4}} - 1.$$

r= 0.0689 = 6.89%.

A is incorrect because this is the average of the annual coupon rate and the prevailing interest rate.

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Discussion

What do you want to discuss... (required)

MH

In Q3 of the practice questions, why are we using N=2 instead of N=3 to calculate the PV which equals -103.3853. I used N=3 to calc the coupon FV, but then have to change it to N=2 to calc the PV for the bond itself

Created 2 months ago by Melik Homer — 0 replies | Last Activity: 2 months ago

Reply to this Comment

ST

In question set 4) is 5.23% not the annualised return, not total return? for total return would we not omit the ^(1/T) ??

Wondering if it is a typo or my understanding of the teminology is wrong

Created 4 months ago by Samuel Twhigg — 1 reply | Last Activity: 3 months ago

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DL

for example 4, = –PV(0.072, 6, 6.2, 100, 0) = 95.263, why did we change it to N = 6? shouldn't it be N = 4? is this a typo?

Created 6 months ago by Dylan Lim Fung Liong Kee — 2 replies | Last Activity: 5 months ago

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AC

I think there is a mistake in the knowledge check question 1. The bond is at par so YTM = Coupon (6.2% on both), then the coupon fall at 5.2% and the YTM remain the same. On financial calculator i got FV (10 N ; 5.2 I/Y ; 6.2 PMT; 0 PV; CPT FV) = 78.71, it should be FV (10 N ; 6.2 I/Y ; 5.2 PMT; 0 PV; CPT FV), confusing

Created 6 months ago by Armand Charles-Edouard Ngoudjo — 2 replies | Last Activity: 5 months ago

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MS

Why do we take the principal repayment of 100$ in calculating the present value of the bond when sold before maturity? As we read in earlier lesson, the present value is calculated by using FV=Principal+final_coupon.

Created 7 months ago by Muhammad Saad Khan Tareen — 1 reply | Last Activity: 7 months ago

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